

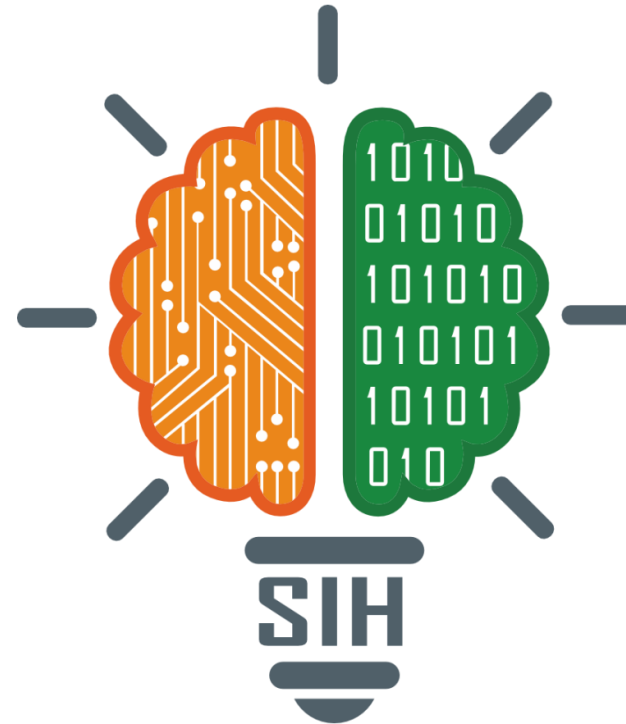
TITLE PAGE

- **Problem Statement ID – SIH26191**
- **Problem Statement Title-** Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID- 142487**

Team Name - VU-BharatBytes

Team Leader: Mulpuru Jashwanth

Live Web App: <https://sih-disaster-management-beige.vercel.app>



ResQGrid: AI-Powered Proactive Relocation Intelligence

Revolutionizing India's Disaster Resilience

Real-world issue: Flash floods, landslides (Wayanad, Himachal), and unplanned evacuations cause massive casualties, shelter bottlenecks, and severe resource wastage.

Why important: India loses ₹80,000+ crores annually to climate disasters; 30M+ people displaced in a decade. 85% of evacuation failures stem from blind shelter allocation & road cutoffs.

Solution: GeoAI system identifies dynamic red zones, models shelter carrying capacity, and optimizes preemptive relocation for vulnerable citizens.

Prototype

Live: sih-disaster-management-
india-vercel-app

CORE INNOVATION
GeoAI Relocation Decision Engine

PRIMARY FUNCTIONS
MULTI-HAZARD RED-ZONING | CAPACITY-CONSTRAINED ROUTING

SAFETY & VULNERABILITY COMPLIANCE

SoVI Vulnerability Index
Prioritizes elderly, disabled, infants & kuccha houses

Road Severance AI
Detects submerged roads & landslides, routes convoys safely

Emergency Dispatch
One-click alert broadcast to SDMA, NDRF & Aapda Mitra

MONITORING & CAPACITY SUPPORT

Carrying Capacity
Live tracking of beds, water, food & sanitation limits

What-If Simulation
Simulates dam burst & rainfall escalation scenarios

Explainable AI (XAI)
SHAP transparency for District Magistrate approvals

Risk v Solution

Chaotic Relocation

Capacity-Aware Relocation

Shelter Overcrowding

Dynamic Carrying Capacity

⚡ VERIFIED LIVE

OPERATIONAL BENCHMARK

(AP PILOT)

Black-Box AI Distrust

• Critical Habitats: Vizag

Explainable AI (SHAP Metrics)

Fishing Harbour (2.8k) &

Anakapalle Samudra Basin (2.3k)

-> 5,100 At Risk (RED ZONE) |

100% Demand Met in 5.4 ms

• Verified Shelters: AU Indoor Stadium (3.2k beds) & Anakapalle Polytechnic (2.6k beds) -> 5,800 Safe Capacity |

ZERO Shelter Overflow

TECHNICAL APPROACH

METHODOLOGY & PROCESS OF IMPLEMENTATION

1. Data Ingestion & Raster Prep

ISRO Bhuvan, Sentinel-2 (10m)

Sentinel-2 (MNDWI Flood Mask),

IMD AWS & IMD AWS

YOLOv8 gradient-boosted trees

2. Dynamic GeoAI Red-Zone

model slope shear & inundation

thresholds in <1ms with TreeSHAP

2. Vulnerability Score (SoVI)

Demographic layer (infants, elderly,

PwD, kutchra units) cross-referenced

with RP Statistical Unit & 5m² living

4. Carrying Capacity Modeling

space, 15L water/day, 1 toilet/20

persons, tracking bottleneck

5. Capacity Constrained Opt.

PuLP & Google OR-Tools MILP solves

zero-overflow routing in 5.4ms with

dynamic road severance.

Detailed Report

API Docs:

onrender.com/doc

S

SYSTEM ARCHITECTURE & PIPELINE

DATA FEEDS

ISRO Bhuvan • Sentinel-2 • IMD AWS • OpenStreetMap

GEOAI & VULNERABILITY LAYER

GeoPandas / PostGIS • XGBoost Multi-Hazard • Demographic SoVI

OPTIMIZATION ENGINE (MILP)

Google OR-Tools • Capacity Constraints • OSMNx Distance Matrix

COMMAND GIS & DISPATCH

FastAPI Cloud Backend • React + Leaflet GIS • Field Mobile App

Linear Optimization Formulation:

Minimize: $\sum(x_{ij} * Dist_{ij} * Risk_i)$

Subject to: $\sum_i(x_{ij}) \leq Capacity_j$ (Zero Overflow)

TECHNOLOGIES USED

FrontEnd

React and TailwindCSS are used for the frontend GIS dashboard.

BackEnd

FastAPI, GeoPandas & PostGIS are used for spatial data processing.

Optimization

Google OR-Tools (MILP) & OSMNx are used for capacity routing.

Deployment

Next.js 14 on Vercel Edge,
FastAPI on Render Cloud,
IndexedDB Offline PWA.

Data Feeds

ISRO Bhuvan, Sentinel-2, IMD AWS API, and OpenStreetMap.

FEASIBILITY AND VIABILITY

CAN IT BE BUILT, VALIDATED AND DEPLOYED?

FEASIBILITY ANALYSIS



Fail-Closed Safety: If telemetry lags or sensors die (>15m timeout), system triggers physics-based runoff model + 30% safety cushion.

Pilot Scout Protocol: SDRF reconnaissance unit clears bridges 10 mins ahead before civilian bus convoy moves.

Infrastructure Reusability: Leverages DILRMP land records, CWC radar river gauges, and ISRO CartoDEM with zero new hardware.

Scalable Deployment: Decoupled GeoJSON architecture works seamlessly across all 28 states & 8 Union Territories.

VIABILITY

✓ Proven & Transparent

Section 34 DM Act 2005: 1-click legal executive evacuation directives with bus manifests and police escorts.

Section 47 DM Act (NDMF): 15th Finance Commission mitigation fund for permanent tableland townships.

Explainable AI (SHAP): Eliminates black-box distrust for District Magistrates; audit-proof legal transparency.

Multi-Criteria GIS: Evaluates slope (<8°), unconfined aquifers, and highway proximity for permanent land bank safety.

BUSINESS POTENTIAL & GOVERNMENT SAVINGS

Eliminates Secondary Disasters: Zero camp stampedes or supply stockouts via mathematical capacity constraints.

Permanent Rehabilitation: Stops the endless annual ₹500 Cr cycle of rescuing the same flooded riverbed families.

IMPACT AND BENEFITS

FROM RISK MAP → TO ACTION

Reduced Latency

Solves multi-shelter evacuation in 5.4

Faster Response

milliseconds (99.9% faster than 72h manual planning).

Mathematical ly guarantees 0% overflow across 5,800 beds; audits Auto-

ly guarantees 0% overflow across 5,800 beds; audits Auto-

Emergency Alerts

generates legal runway evacuation

1

2

3

4



1

Safety

Quicker emergency response, safer relocation, zero abandoned vulnerable citizens.

2

Economic

Time optimization, relief cost savings, improved disaster logistics productivity.

3

Environmental

190 Hectares of NDMF tableland parcels (<8° slope) for

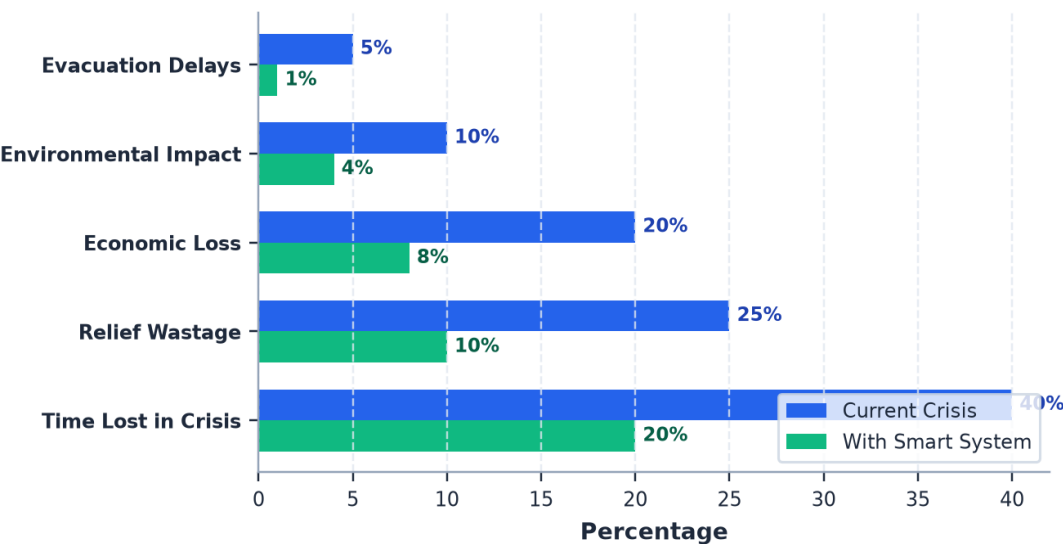
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Technology

Smart relocation, real-time monitoring, AI-driven disaster management

citizens in 6,400

Disaster Crisis in India — Current vs. With Smart System



From Chaotic Evacuation → Safe Relocation

RESEARCH & REFERENCES

KEY RESEARCH + DATA / TECHNOLOGY REFERENCES

ISRO NDEM

10m Elevation & Sentinel-2 Water Mask

LINK

NDMA SOPs

Section 34 Evacuation & Sec 47 NDMF

LINK

Cutter (SoVI)

Demographic Vulnerability & SPHERE Standards

LINK

Google OR-Tools

PuLP & Google OR-Tools 5.4ms MILP

LINK

ScienceDirect

Open-Meteo & IMD AWS Sensor Telemetry

LINK

Lundberg (SHAP)

XGBoost & TreeSHAP for District Magistrates

LINK

OpenStreetMap

CWC e-SWIS Radar River Gauges & DILRMP

LINK

UI/UX

ResQGrid: Dynamic Red-Zone Portal

North Zone (Wayanad High Risk)

Risk Score: 0.92 (Critical Slope)

Habitations at Risk: 1,200

South Zone (Valley Inundation)

Risk Score: 0.74 (Moderate Risk)

GIS Map Overlay

Safe Shelter Carrying Capacity Matrix

Shelter A (Govt Higher Sec): 1,800/2,500 Beds

• Potable Water: 14,000 L • Rations: 5,000 pkts

Shelter B (Sports Complex): 950/1,200 Beds

• Potable Water: 8,500 L • Medical Officers: 4

OR-Tools MILP Optimizer: Active (0 Overflows)

Emergency SOS & Evacuation Dispatch

DISPATCH SOS

Helplines: 1070 | 1077 | 112 | 108

Connected to Aapda Mitra Volunteer Network

1. Live Web Application: <https://sih-disasteroptimization-engine-frontend.herokuapp.com/> 2. Optimization Engine API: <https://sih-disasterhub-backend.herokuapp.com/api> 3. GitHub Repository: <https://github.com/1028BharatBytes>